

FIGURE 6.1

Number of plants observed in tracts at oil spill and control sites. The data are displayed in stem-and-leaf plots

Control Tracts				Oil Spill Tracts			
Mean:	38.48	000	0	Mean:	26.93		
Median:	41.50	7	0	Median:	26.00		
St. Dev:	16.37	1	1	St. Dev:	9.88		
n:	40	6	1	n:	40		
		4	2				
		9	2				
		0	3				
		55678	3				
		000111222233	4				
		57	4				
		0112344	5				
		67789	5				

TABLE 6.1

Summary statistics for oil spill data

Descriptive Statistics						
Variable	Site Type	N	Mean	Median	Tr. Mean	St. Dev.
No. plants	Control	40	38.48	41.50	39.50	16.37
	Oil spill	40	26.93	26.00	26.69	9.88
Variable	Site Type	SE Mean	Minimum	Maximum	Q1	Q3
No. plants	Control	2.59	0.00	59.00	35.00	51.00
	Oil spill	1.56	5.00	52.00	22.00	33.75

summary statistics given in Table 6.1. A boxplot of the data displayed in Figure 6.2 indicates that the control sites have a somewhat greater plant density than the oil spill sites. From the summary statistics, we have that the average flora density in the control sites is $\bar{y}_{\text{Con}} = 38.48$ with a standard deviation of $s_{\text{Con}} = 16.37$. The sites within the spill region have an average density of $\bar{y}_{\text{Spill}} = 26.93$ with a standard deviation of $s_{\text{Spill}} = 9.88$. Thus, the control sites have a larger average flora density and a greater variability in flora density than the sites within the spill region. Whether these observed differences in flora density reflect similar differences in

FIGURE 6.2

Number of plants observed in tracts at control sites (1) and oil spill sites (2)

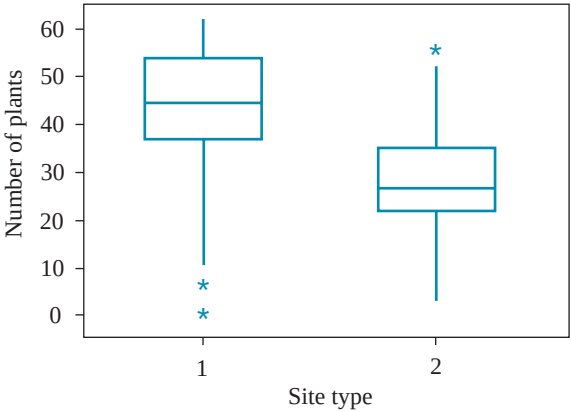
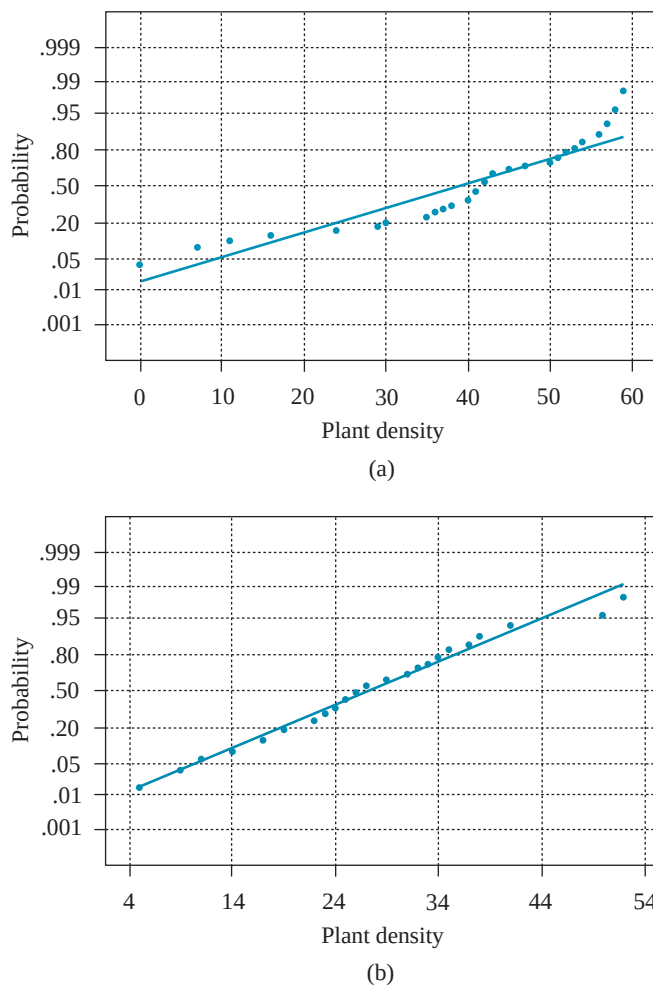


FIGURE 6.6

(a) Normal probability plot for control sites. (b) Normal probability plot for oil spill sites



t procedures will be the most appropriate inference procedures. The sample data yielded the following values:

Control Plots	Oil Spill Plots
$n_1 = 40$	$n_2 = 40$
$\bar{y}_1 = 38.48$	$\bar{y}_2 = 26.93$
$s_1 = 16.37$	$s_2 = 9.88$

The research hypothesis is that the mean plant density for the control plots exceeds that for the oil spill plots. Thus, our approximate t test is set up as follows:

$$H_0: \mu_1 \leq \mu_2 \text{ versus } H_a: \mu_1 > \mu_2$$

That is,

$$H_0: \mu_1 - \mu_2 \leq 0$$

$$H_a: \mu_1 - \mu_2 > 0$$

$$\text{T.S.: } t' = \frac{(\bar{y}_1 - \bar{y}_2) - D_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(38.48 - 26.93) - 0}{\sqrt{\frac{(16.37)^2}{40} + \frac{(9.88)^2}{40}}} = 3.82$$

To compute the rejection region and p -value, we need to compute the approximate df for t' :

$$c = \frac{\frac{s_1^2/n_1}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \frac{(16.37)^2/40}{(16.37)^2/40 + (9.88)^2/40} = .73$$

$$\text{df} = \frac{(n_1 - 1)(n_2 - 1)}{(1 - c)^2(n_1 - 1) + c^2(n_2 - 1)} = \frac{(39)(39)}{(1 - .73)^2(39) + (.73)^2(39)}$$

$$= 64.38, \text{ which is rounded to } 64$$

Table 2 in the Appendix does not have df = 64, so we will use df = 60. In fact, the difference is very small when df becomes large: $t_{.05} = 1.671$ and 1.669 for df = 60 and 64, respectively.

R.R.: For $\alpha = .05$ and df = 60, reject H_0 if $t' > 1.671$.

Since $t' = 3.82$ is greater than 1.671, we reject H_0 . We can bound the p -value using Table 2 in the Appendix with df = 60. With $t' = 3.82$, the level of significance is $p\text{-value} < .001$. Thus we can conclude that there is significant ($p\text{-value} < .001$) evidence that μ_1 is greater than μ_2 . Although we have determined that there is a statistically significant difference between the mean plant densities at the control and oil spill sites, the question remains whether these differences have *practical* significance. We can estimate the size of the difference in the means by placing a 95% confidence interval on $\mu_1 - \mu_2$.

The appropriate 95% confidence interval for $\mu_1 - \mu_2$ is computed by using the following formula with df = 64, the same as was used for the R.R.

$$(\bar{y}_1 - \bar{y}_2 \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad \text{or} \quad (38.48 - 26.93) \pm 2.0 \sqrt{\frac{(16.37)^2}{40} + \frac{(9.88)^2}{40}}$$

$$\text{or} \quad 11.55 \pm 6.05$$

Thus, we are 95% confident that the mean plant densities differ by an amount between 5.5 and 17.6. The plant scientists would then evaluate whether a difference in this range is of practical importance.

Reporting Conclusions We would need to write a report summarizing our findings from the study. The report would include the following items:

1. Statement of the objective for the study
2. Description of the study design and data collection procedures
3. Numerical and graphical summaries of the data sets
 - Table of means, medians, standard deviations, quartiles, range
 - Boxplots
 - Stem-and-leaf plots
4. Description of all inference methodologies:
 - Approximate t tests of differences in means
 - Approximate t -based confidence interval on population means
 - Verification that all necessary conditions for using inference techniques were satisfied using boxplots, normal probability plots